



2 Real Analysis 2023
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Deferred Final Exam
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Instructions:

- This question paper consists of **9 pages** (including this one).
- You have 120 minutes to answer the questions.
- Answer all questions in the corresponding boxes.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 70 points = 100% (74 points available).

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Problem 1

(a) Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2 - x^2$.

(i) Determine, without proof, the image $f(A)$ of the set $A = [-1, 2]$. (1 points)

Solution. $f(A) = [-2, 2]$.

(ii) Determine, without proof, the pre-image $f^{-1}(B)$ of the set $B = [0, \infty)$. (1 points)

Solution. $f^{-1}(B) = [-\sqrt{2}, \sqrt{2}]$.

(b) Consider the set $S = (-1, 0) \cup \left\{ \frac{n-1}{n} : n \in \mathbb{N} \right\}$.

(i) Determine, without proof, $\sup S$. (1 points)

Solution. $\sup S = 1$.

(ii) Determine, without proof, the set S' of all limit points of S . (1 points)

Solution. $S' = [-1, 0] \cup \{1\}$.

(c)

(i) Give, without proof, an example of a countable set. (1 points)

Solution. For example, $\mathbb{N}$ is a countable set.

(ii) Give, without proof, an example of an uncountable set. (1 points)

Solution. For example, $\mathbb{R}$ is an uncountable set.

Problem 2

For each of the following series, determine whether it is absolutely convergent, conditionally convergent or divergent. Give a justification.

$$(a) \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)^2 + n} \quad (4 \text{ points})$$

Solution. For any $n \in \mathbb{N}$ we have

$$\left| \frac{(-1)^n}{(2n)^2 + n} \right| = \frac{1}{4n^2 + n} \leq \frac{1}{4n^2}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges (this is a convergent p -series with $p = 2$), and by the Arithmetic Limit Theorem the series $\sum_{n=1}^{\infty} \frac{1}{4n^2}$ converges. By the Comparison Test, the series $\sum_{n=1}^{\infty} \frac{1}{4n^2 + n}$ converges.

We conclude that the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)^2 + n}$ converges absolutely.

$$(b) \sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2} \quad (4 \text{ points})$$

Solution. We apply the Ratio Test:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} = \lim_{n \rightarrow \infty} \frac{(2n+2)!}{((n+1)!)^2} \frac{(n!)^2}{(2n)!} = \lim_{n \rightarrow \infty} \frac{(2n+2)(2n+1)}{(n+1)(n+1)} \\ &= \lim_{n \rightarrow \infty} \frac{\left(2 + \frac{2}{n}\right) \left(2 + \frac{1}{n}\right)}{\left(1 + \frac{1}{n}\right) \left(1 + \frac{1}{n}\right)} = 4 > 1. \end{aligned}$$

We conclude that the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)^2 + n}$ diverges.

$$(c) \sum_{n=1}^{\infty} (-1)^n \left(1 + \frac{1}{n}\right)^n \quad (2 \text{ points})$$

Solution. We have

$$\lim_{n \rightarrow \infty} \left| (-1)^n \left(1 + \frac{1}{n}\right)^n \right| = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

It follows that the terms of the series $(-1)^n \left(1 + \frac{1}{n}\right)^n$ do not converge to zero as $n \rightarrow \infty$.

We conclude that the series $\sum_{n=1}^{\infty} (-1)^n \left(1 + \frac{1}{n}\right)^n$ diverges.

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(d) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n(n+1)}}$ (6 points)

Solution. First we consider the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}}$. For any $n \in \mathbb{N}$ we have

$$\frac{1}{\sqrt{n(n+1)}} \geq \frac{1}{\sqrt{n(n+n)}} = \frac{1}{\sqrt{2n^2}} = \frac{1}{\sqrt{2}n}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges (this is a divergent harmonic series). By the Arithmetic Limit Theorem, the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2}n}$ diverges. By the Comparison Test, the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}}$ diverges. Thus, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n(n+1)}}$ is not absolutely convergent.

The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n(n+1)}}$ is an alternating series. We have that $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n(n+1)}} = 0$ and $\frac{1}{\sqrt{n(n+1)}} \geq \frac{1}{\sqrt{(n+1)(n+2)}}$, $n \in \mathbb{N}$. By the Alternating Series Test, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n(n+1)}}$ converges. The final answer is that the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n(n+1)}}$ converges conditionally.

Problem 3

Read the following proposition and an outline of its proof.

Proposition. Let $A \subseteq \mathbb{R}$ be a nonempty set that is bounded above and closed. Then $\sup A \in A$.

Proof. Let $a = \sup A$. Assume, by contradiction, that $a \notin A$. Then for any $\varepsilon > 0$ there is a point $a_\varepsilon \in A$ such that $a - \varepsilon < a_\varepsilon < a$. Thus, a is a limit point of A , which is a contradiction. The contradiction shows that $a \in A$.

Answer the following questions:

- (a) Define what it means that A is bounded above. Define an upper bound for A . (2 points)

Solution. A is bounded above if there is $M \in \mathbb{R}$ such that $x \leq M$ for any $x \in M$. Any M with this property is called an upper bound for A .

- (b) Define $\sup A$. (3 points)

Solution. $a = \sup A$ if the following two conditions are fulfilled:

- 1) a is an upper bound for A ,
- 2) if M is any upper bound for A , then $a \leq M$.

- (c) Which statement from the lecture guarantees that $\sup A$ exists? Name this statement and state it clearly. (2 points)

Solution. This is the Completeness Axiom: Every nonempty subset of $\mathbb{R}$ which is bounded above has a supremum.

- (d) Explain why $a_\varepsilon \in A$ such that $a - \varepsilon < a_\varepsilon < a$ exists. (4 points)

Solution. Since a is the least upper bound for A and $a - \varepsilon < a$, we conclude that $a - \varepsilon$ is not an upper bound for A . Therefore there exists a point in A , say $a_\varepsilon \in A$, with $a_\varepsilon > a - \varepsilon$. On the other hand, $a_\varepsilon \leq a$, because a is an upper bound for A and $a_\varepsilon \in A$. But we have by assumption that $a \notin A$. It follows that $a_\varepsilon < a$.

- (e) Give a definition of a limit point of A . (2 points)

Solution. x is a limit point of A , if every ε -neighborhood $(x - \varepsilon, x + \varepsilon)$, $\varepsilon > 0$, contains a point from $A \setminus \{x\}$.

- (f) Give a definition of closed set. (2 points)

Solution. A set A is closed if A contains all its limit points.

- (g) Explain how we obtain a contradiction. (4 points)

Solution. Since for any $\varepsilon > 0$ the neighborhood $(a - \varepsilon, a + \varepsilon)$ contains a point $a_\varepsilon \in A \setminus \{a\}$, we conclude that a is a limit point of A . Since A is closed, we must have $a \in A$. But this contradicts our assumption that $a \notin A$.

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Problem 4

Suppose that $(a_n)_{n=1}^{\infty}$ is a sequence with $\lim_{n \rightarrow \infty} a_n = 7$. Prove by definition that

$$\lim_{n \rightarrow \infty} \frac{a_n + 2}{a_n - 3} = \frac{9}{4}.$$

(8 points)

Solution. We have

$$\left| \frac{a_n + 2}{a_n - 3} - \frac{9}{4} \right| = \left| \frac{4(a_n + 2) - 9(a_n - 3)}{4(a_n - 3)} \right| = \frac{|-5a_n + 35|}{4|a_n - 3|} = \frac{5|a_n - 7|}{4|a_n - 3|}.$$

By the definition of the limit $\lim_{n \rightarrow \infty} a_n = 7$, there is $N_1 \in \mathbb{N}$ such that

$$|a_n - 7| < 1 \quad \text{for all } n \geq N_1.$$

For such n we have the following chain of equivalent inequalities: $|a_n - 7| < 1 \iff -1 < a_n - 7 < 1 \iff 6 < a_n < 8 \iff 3 < a_n - 3 < 5$, the latter inequality implying $|a_n - 3| > 3$. Thus, if $n \geq N_1$, then

$$\left| \frac{a_n + 2}{a_n - 3} - \frac{9}{4} \right| = \frac{5|a_n - 7|}{4|a_n - 3|} < \frac{5}{12}|a_n - 7| < |a_n - 7|.$$

Given $\varepsilon > 0$, there is $N_\varepsilon \in \mathbb{N}$ such that

$$|a_n - 7| < \varepsilon \quad \text{for all } n \geq N_\varepsilon.$$

Put $N = \max\{N_1, N_\varepsilon\}$. Then

$$\left| \frac{a_n + 2}{a_n - 3} - \frac{9}{4} \right| < |a_n - 7| < \varepsilon \quad \text{for all } n \geq N.$$

By the definition of the limit, $\lim_{n \rightarrow \infty} \frac{a_n + 2}{a_n - 3} = \frac{9}{4}$.

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Problem 5

Consider the function $f(x) = xe^x$ on the interval $[0, 1]$.

(a) Use the Mean Value Theorem to prove that $|f(x_1) - f(x_2)| \leq 2e|x_1 - x_2|$ for $x_1, x_2 \in [0, 1]$. (5 points)

Solution. By the Mean Value Theorem, for any $x_1, x_2 \in [0, 1]$ there is t between x_1 and x_2 such that

$$f(x_1) - f(x_2) = f'(t)(x_1 - x_2).$$

This implies

$$|f(x_1) - f(x_2)| = |f'(t)| \cdot |x_1 - x_2|.$$

We have $f'(t) = e^t + te^t = (1 + t)e^t$. For $t \in [0, 1]$ we have $0 \leq f'(t) \leq (1 + 1)e^1 = 2e$. Therefore, $|f'(t)| \leq 2e$ for $t \in [0, 1]$, and

$$|f(x_1) - f(x_2)| \leq 2e|x_1 - x_2|$$

for any $x_1, x_2 \in [0, 1]$, as desired.

(b) Prove by definition that f is uniformly continuous on $[0, 1]$. (5 points)

Solution. Given $\varepsilon > 0$, take $\delta = \frac{\varepsilon}{2e} > 0$. Then for any $x_1, x_2 \in [0, 1]$,

$$|x_1 - x_2| < \delta \implies |f(x_1) - f(x_2)| \leq 2e|x_1 - x_2| < 2e\delta = 2e \frac{\varepsilon}{2e} = \varepsilon.$$

It follows that f is uniformly continuous on $[0, 1]$.

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Problem 6

Consider a sequence of functions $(f_n)_{n=1}^\infty$, where $f_n : [0, 1] \rightarrow \mathbb{R}$, $f_n(x) = \frac{nx}{1 + 2nx}$.

(a) Determine the pointwise limit f of the sequence $(f_n)_{n=1}^\infty$ on $[0, 1]$. (5 points)

Solution. If $x = 0$, then $f_n(0) = 1$ for all $n \in \mathbb{N}$, and thus $\lim_{n \rightarrow \infty} f_n(0) = 1$.

If $x \neq 0$, then

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{nx}{1 + 2nx} = \lim_{n \rightarrow \infty} \frac{x}{\frac{1}{n} + 2x} = \frac{x}{2x} = \frac{1}{2}.$$

Thus, the pointwise limit of the sequence $(f_n)_{n=1}^\infty$ is

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 1, & x = 0, \\ \frac{1}{2}, & x \in (0, 1]. \end{cases}$$

(b) Does $(f_n)_{n=1}^\infty$ converge to f uniformly on $[0, 1]$? Justify your answer. (3 points)

Solution. For each $n \in \mathbb{N}$, the function f_n is continuous on $[0, 1]$. However, the limit function $f = \lim_{n \rightarrow \infty} f_n$ is discontinuous on $[0, 1]$ (more precisely, it is discontinuous at 0). We know that the uniform limit of a sequence of continuous functions is continuous. It follows that $(f_n)_{n=1}^\infty$ does not converge to f uniformly on $[0, 1]$.

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Problem 7

Consider the series $\sum_{n=0}^{\infty} \frac{\sin nx}{2^n}$, $x \in \mathbb{R}$.

(a) Show that this series converges absolutely and uniformly on $\mathbb{R}$. (4 points)

Solution. We have for all $x \in \mathbb{R}$ and all $n = 0, 1, 2, \dots$

$$\left| \frac{\sin nx}{2^n} \right| \leq \frac{1}{2^n} = M_n,$$

and the series $\sum_{n=0}^{\infty} M_n = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$ converges (it is a converging geometric series).

By the Weierstrass M-Test, the series $\sum_{n=0}^{\infty} \frac{\sin nx}{2^n}$ converges absolutely and uniformly on $\mathbb{R}$.

(b) Justify that the function $g(x) = \sum_{n=0}^{\infty} \frac{\sin nx}{2^n}$ is continuous on $\mathbb{R}$. (3 points)

Solution. Put $s_N(x) = \sum_{n=0}^N \frac{\sin nx}{2^n}$, $N \in \mathbb{N}$. Each function $s_N(x)$ is continuous on $\mathbb{R}$, being a trigonometric polynomial. By part (a), s_N converges to g uniformly on $\mathbb{R}$ as $N \rightarrow \infty$. We know that a uniform limit of a sequence of continuous functions is continuous. It follows that g is continuous on $\mathbb{R}$,